

Evaluating Limits



Direct substitution and algebraic techniques

1 Evaluate each limit:

a $\lim_{x \rightarrow 1} (3x^2 - 2x + 5)$

b $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

c $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$

2 Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x^2 - 9}$.

Trigonometric and special limits

3 Using $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, evaluate:

a $\lim_{x \rightarrow 0} \frac{\sin 5x}{x}$

b $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$

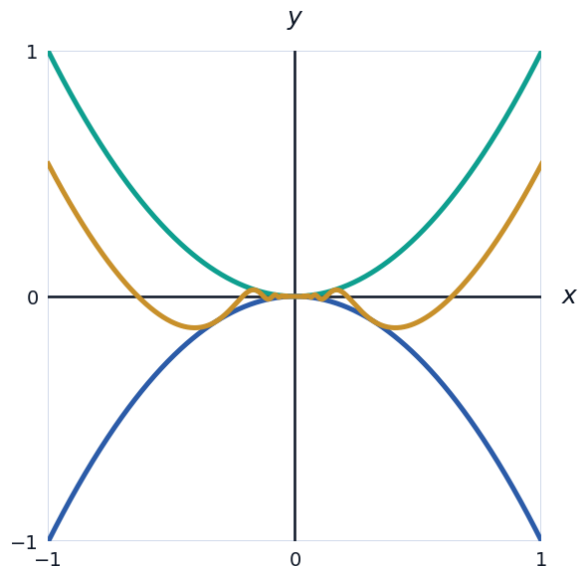
c $\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 5x}$

4 Evaluate $\lim_{x \rightarrow \infty} \frac{5x^3 - 2x + 1}{2x^3 + 7}$.

Limits that fail to exist and the squeeze theorem

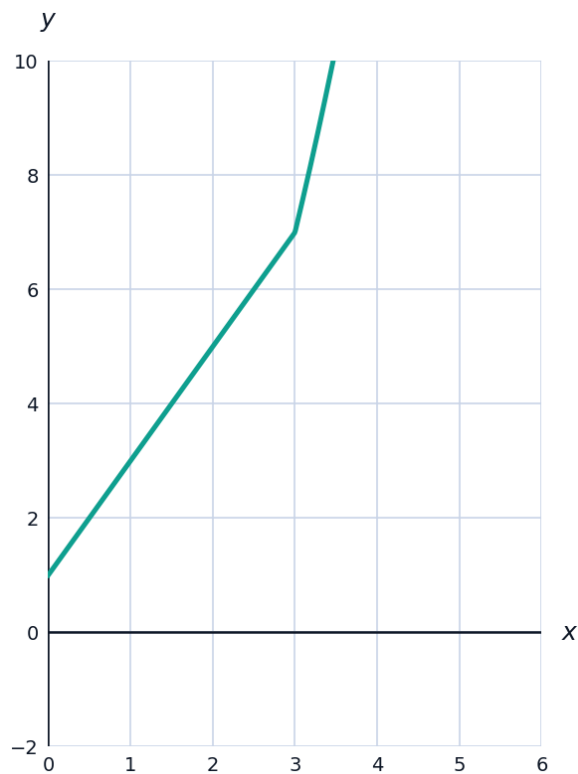
5 Explain why $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$ does not exist.

- 6 Use the squeeze theorem to evaluate $\lim_{x \rightarrow 0} x^2 \cos\left(\frac{1}{x}\right)$.



Limits with absolute value and piecewise functions

- 7 Evaluate $\lim_{x \rightarrow 3} |x-3| \cdot \frac{1}{x}$.
- 8 For $f(x) = 2x + 1$ if $x < 3$, and $f(x) = x^2 - 2$ if $x \geq 3$, evaluate $\lim_{x \rightarrow 3^-} f(x)$ and $\lim_{x \rightarrow 3^+} f(x)$, and state whether $\lim_{x \rightarrow 3} f(x)$ exists.

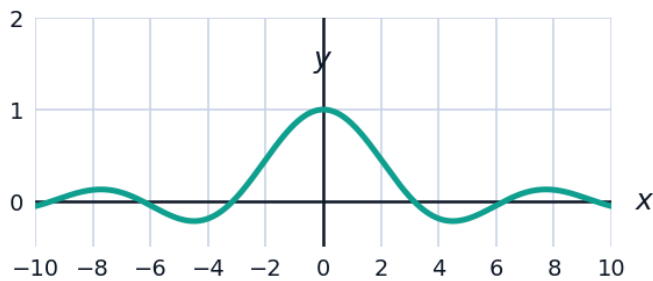


More algebraic limit techniques

- 9 Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$ by factoring with the difference-of-cubes formula.
- 10 Evaluate $\lim_{h \rightarrow 0} \frac{(2+h)^2 - 4}{h}$.
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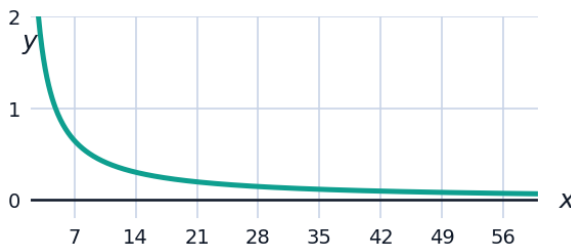
Visualizing a limit numerically and graphically

- 11 The function $f(x) = \frac{\sin x}{x}$ (undefined at $x = 0$) is graphed below. Use the graph to estimate $\lim_{x \rightarrow 0} f(x)$.



Limits at infinity with different degree relationships

- 12 Evaluate each limit at infinity, based on the relative degrees of numerator and denominator:



- a $\lim_{x \rightarrow \infty} \frac{4x+3}{x^2-1}$ (lower degree numerator)
- b $\lim_{x \rightarrow \infty} \frac{x^3-1}{2x^2+5}$ (higher degree numerator)
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Limits requiring conjugate multiplication

- 13 Evaluate $\lim_{x \rightarrow 9} \frac{x-9}{\sqrt{x}-3}$ by rationalizing the denominator.
- 14 Evaluate $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 3x} - x)$ by rationalizing.

Piecewise limits requiring a parameter

- 15 For $f(x) = kx + 1$ if $x \leq 2$, and $f(x) = x^2 - 3$ if $x > 2$, find k so that $\lim_{x \rightarrow 2} f(x)$ exists.

Limits involving factorable higher-degree polynomials

- 16 Evaluate $\lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1}$ by factoring the numerator as a difference of squares twice.
- 17 Evaluate $\lim_{x \rightarrow -2} \frac{x^3 + 8}{x + 2}$ using the sum-of-cubes factoring pattern.
- 18 Evaluate $\lim_{x \rightarrow 0} \frac{(3+x)^2 - 9}{x}$.